Multitask Fine-Grained Feature Mining for Multilabel Remote Sensing Image Classification

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***Abstract*— Multilabel remote sensing image classification can provide comprehensive object-level semantic descriptions of remote sensing images. However, most existing methods cannot fully mine the fine-grained features of images and labels, resulting in low classification accuracy. To address this issue, we propose a novel multitask framework for multilabel remote sensing image classification. The framework establishes the class-specific feature extraction as a binary classification auxiliary task to assist the main multilabel classification task, which can improve the model's local and global feature extraction ability. Meanwhile, the framework updates the label correlation graph using the graph transformer layer to accurately identify label node pairs with potential correlation, which effectively mines the correlation of multiple labels to generate more accurate label co-occurrence embedding for image label prediction. Experimental results on UCM, AID, and DFC15 multilabel datasets show that the proposed method outperforms existing state-of-the-art methods.**

***Index Terms*— Fine-grained feature, multilabel image classification, multitask framework, remote sensing.**

I. Introduction

With the rapid development of remote sensing technology, the clarity of remote sensing images has been dramatically improved, and high-resolution remote sensing images have reached the practical stage. They are widely used in many fields, such as land use determination [1], [2], [3], ecological detection [4], [5], and traffic control [6]. Among these applications, remote sensing image classification is one of the fundamental vision tasks. It aims to extract semantic-level information from images and assign specific labels to them, assisting in the implementation of various remote sensing tasks that have attracted extensive attention from researchers [7], [8], [9].

In the past, most remote sensing images were annotated with a single scene label, which refers to the overall visual understanding of the remote sensing image content [10], [11]. However, existing methods are severely limited by their reliance on predetermined scene labels, which perform poorly in new scenes other than the predetermined ones. To address this issue, the remote sensing knowledge graph is used to enhance the semantic understanding and correlation reasoning ability of the model, enabling the model to identify unseen remote sensing image scenes to achieve zero-shot scene classification [12]. Furthermore, remote sensing large models (e.g., SkySense [13]) have recently been proposed to perform multiple tasks simultaneously, such as scene classification and object detection, which allow for a more comprehensive description and understanding of remote sensing images.

However, noted that high-resolution remote sensing images contain multiple different semantic objects and complex spatial geometric information. The conventional single-label annotation focuses on understanding remote sensing images from a macro perspective, and it is impossible to fully describe each semantic object in remote sensing images. To address this issue, multilabel image annotation was proposed to make the description of images with specific object-level labels, which enables a more fine-grained understanding of the remote sensing image [14]. As shown in Fig. 1, for the same image, multilabel annotation can describe more diverse objects in the image, such as grass, trees, and water, rather than just identifying the river in the image as with single-label annotation. Accordingly, multilabel annotation is necessary for complex tasks based on remote sensing images such as image classification and image retrieval. The multilabel remote sensing image classification task aims at extracting comprehensive features of the remote sensing image to match multiple object-level labels, which still remains a challenge.

One major challenge is how to extract high-level semantic features from remote sensing images. Traditional methods mainly rely on hand-crafted features and mid-level semantic models, which only obtain shallow local features of images, leading to the limited performance of classification [15], [16], [17]. With the rise of deep learning, the convolutional neural network (CNN) has been widely used for various computer

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(a) (b)

Fig. 1. Illustration of remote sensing images with single scene labels and multiple object labels. (a) Single scene label: river; multiple object labels: bare-soil, buildings, cars, dock, field, grass, pavement, ship, trees, and water. (b) Single scene label: river; multiple object labels: bare-soil, chaparral, grass, trees, and water.

vision tasks with excellent image feature extraction abilities to obtain high-level semantic features of images. However, as the depth of the CNN network increases, many detailed features become progressively blurry after multilayer convolution, which easily leads to the missing of small objects in remote sensing images [18], [19], [20]. Meanwhile, most of the remote sensing images contain wide natural scenes and artificial buildings, which have strong interclass similarities [21], [22]. As a result, CNN-based classifiers will easily confuse similar objects and will fail to accurately predict all category labels [21], [23]. Besides CNN, other innovative deep learning models, such as vision transformer (ViT), have been proposed in the field of remote sensing [24]. ViT is based on the self-attention mechanism, which effectively captures the global interaction and exploits long-distance semantic representation by learning the relationship between sequence elements. However, the structural information of the remote sensing image is destroyed during the flattening process to obtain sequence elements. Furthermore, compared with natural image datasets (e.g., MS-COCO [25]), the amount of data available for training in multilabel remote sensing datasets is relatively small, which limits the semantic feature representation ability and generalization performance of the model. Therefore, the direct use of rough high-level features extracted with CNN or ViT cannot satisfy the diverse requirements of multilabel remote sensing image classification.

Another major challenge is how to exploit the complementary information to improve the multilabel image classification accuracy. One solution is to reveal the underlying correlation within different labels [26]. For example, the presence of the "cars" can infer a high probability of the "pavement," while the presence of the "sea" can infer a low probability of the "airplane." To extract the correlation of multiple labels, recurrent neural networks (RNNs) and long–short-term memory (LSTM) are the commonly used models [27], [28]. Besides label correlation, the knowledge graph can also provide supplementary information in multilabel image classification tasks, which utilizes existing human knowledge to supplement label information [29]. However, the chain propagation structure of RNN and LSTM cannot effectively explore the correlation among all labels, which greatly affects the efficiency of label prediction [30]. To overcome this restriction, the graph convolutional network (GCN) was proposed to establish the global correlation between labels by introducing a graph structure among labels [26], [31]. However, the label correlation matrix based on GCN overly relies on label node pairs with strong correlation in the graph while easily ignoring the positive effect of weakly correlated label node pairs, making it difficult to accurately characterize the association between multiple object-level labels in complex scenarios [32], [33].

In summary, existing multilabel classification methods for remote sensing images still suffer from the following open challenges: the relatively small amount of data available for training, objects with different sizes and strong similarity in remote sensing images, and ignorance of weakly correlated label pairs with positive effects in the label correlation graph.

To address the above issues, we propose a novel multitask fine-grained feature mining framework for multilabel remote sensing image classification, which includes the main multilabel image classification task and the binary classification auxiliary task. The framework consists of three modules: an image feature extraction module (IFEM), a label co-occurrence embedding module (LCEM), and a class-specific feature extraction module (CFEM). Specifically, the IFEM fuses multilevel features of remote sensing images and enhances the role of dominant features for the classification task to extract fine-grained features from remote sensing images. The LCEM uses the graph transformer layer to update the initial label correlation graph obtained by counting the probability of co-occurrence of label pairs, which improves the correlation of potentially useful label node pairs in the graph to enable more accurate features matching between the remote sensing image and multiple labels. The CFEM learns class-specific spatial pooling coefficients to generate class-specific features for each class, then establishes the binary classification auxiliary task to assist the main multilabel image classification task. In this way, the binary classification auxiliary task, which uses the same dataset as the main multilabel image classification task, can refine the image feature by providing additional supervision during training. Experimental results on UCM, AID, and DFC15 multilabel datasets show that the proposed method outperforms existing state-of-the-art methods, including ML-GCN [26], ML-CG [29], and SIGNA [34].

The main contributions of this article are as follows.

1) The novel multitask framework for multilabel remote sensing image classification is proposed, which establishes a binary classification auxiliary task using CFEM to assist the main multilabel image classification task. This framework largely improves training efficiency and classification accuracy.
2) The multiscale fusion strategy and the visual attention mechanism are proposed in IFEM to extract fine-grained features from remote sensing images, which can help identify objects with different sizes and distinguish objects with strong similarity in remote sensing images.
3) The graph transformer layer is integrated into the GCN in LCEM to accurately identify label node pairs with potential correlation in the label correlation graph, which generates more accurate label co-occurrence embedding for image label prediction.

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The rest of this article is organized as follows. Section II reviews existing multilabel image classification algorithms in remote sensing. Section III introduces the proposed method in detail. Section IV presents the experimental results and discussions, followed by the conclusion in Section V.

II. Related Work

Due to its practical value, multilabel remote sensing image classification has always been a hot research topic [35]. In this section, we review the related work from the following aspects: multilabel remote sensing image classification, label correlation learning, and multitask learning in remote sensing.

A. Multilabel Remote Sensing Image Classification

The most crucial task of image multilabel classification is to extract suitable semantic features from remote sensing images. Traditional methods of multilabel image classification rely on handcrafted features [16], which are used to describe attributes such as color, texture, and shape of objects. In the early days, researchers mainly combined handcrafted features with traditional machine learning methods for multilabel image classification. Koda et al. [36] introduced the contextual relationship of image region space into a structured support vector machine (SVM) for multilabel image classification. Zeggada et al. [37] used a conditional random field (CRF) model to integrate the spatial correlation between adjacent image patches and the correlation between labels within the same image to improve the performance of multilabel image classification. However, the above methods mainly rely on handcrafted features, they cannot effectively extract high-level semantic features due to the complex spatial structure of remote sensing images, which significantly restricts the classification performance. With the rapid development of deep learning, deep CNNs such as VGGNet [38], ResNet [39], and InceptionNet [40] have powerful feature representation capabilities and are used for multilabel image classification. Zeggada et al. [41] divided remote sensing images into regions of equal size, then extracted their features with CNN, and finally fed these features into a multilabel classification layer with thresholding operation to predict multiple labels for remote sensing images. Gardner and Nichols [23] used pretrained CNN to extract image features and introduced binary cross-entropy (BCE) loss for multilabel image classification. Zhu et al. [18] used a two-branch CNN for joint learning to obtain the multilevel semantic representation of remote sensing images. Tong et al. [42] proposed a network with channel attention and used densely connected CNN layers to extract features at multiple scales and correlate them with each other. However, the above methods only focus on image information and do not consider the correlations between the remote sensing image labels.

B. Label Correlation Learning

The correlations between multiple labels can provide important supplementary information for the multilabel image classification task, which reveals the potential correlation between different labels in a certain image [26]. Introducing the label correlation information into the multilabel classification task of remote sensing images has also proven to be effective.

Hua et al. [2] proposed a CNN-LSTM network, which used CNN to extract image features and send them to a bidirectional LSTM for classification to learn the correlation between labels. Huang et al. [43] used a fully connected layer to extract the label-related information of the label co-occurrence matrix and embed it into the image multiscale attention features. Bazi et al. [44] proposed a two-branch network for multilabel image classification and optimized the loss function related to label discrimination. Lin et al. [29] completed the label information by adding ConceptNet to generate a concept graph and used GNN to extract label correlation information. Liu et al. [34] used semantical interleaving encoding to guide image features into a semantic feature space embedded with label information and utilized channel attention in the semantic feature space to extract more important visual features.

C. Multitask Learning in Remote Sensing

Multitask learning simultaneously uses the same dataset for different tasks to provide additional supervision during the training procedure, which can enhance the feature representation ability of the model to improve the performance of the main task effectively [45].

Kandala et al. [46] proposed a transformer-based model for remote sensing image captioning and multilabel image classification tasks, where the multilabel image classification was used as an auxiliary task to assist the image captioning task, and these two tasks shared the transformer encoder. Bischke et al. [47] introduced a weighted multitask loss based on distance transformation to improve the semantic segmentation prediction of the model and the generalization of the model. Abadal et al. [48] proposed a dual-path network for super-resolution and semantic segmentation of remote sensing images, which has a shared encoder and two dedicated decoders for each task to significantly improve the accuracy of semantic segmentation. Zeng et al. [49] proposed an attention-embedding multitask network for remote sensing scene classification, which uses a coarse-grained classifier and a fine-grained classifier to perform multitask learning to enhance the multigrained feature representation ability of the model. Hang et al. [50] designed a generator network to perform the reconstruction task and the classification task simultaneously to improve the discrimination and generalization ability of the classification task.

III. Methodology

The overall framework of our method is illustrated in Fig. 2, which mainly consists of three modules: the IFEM, the LCEM, and the CFEM. First, we employ a multiscale fusion strategy and the visual attention mechanism to obtain image representation at different scales and highlight the role of dominant features. At the same time, we update the initial graph structure using the graph transformer layer to accurately identify label node pairs with potential correlation in the graph, and use GCN to mine high-order correlation between label

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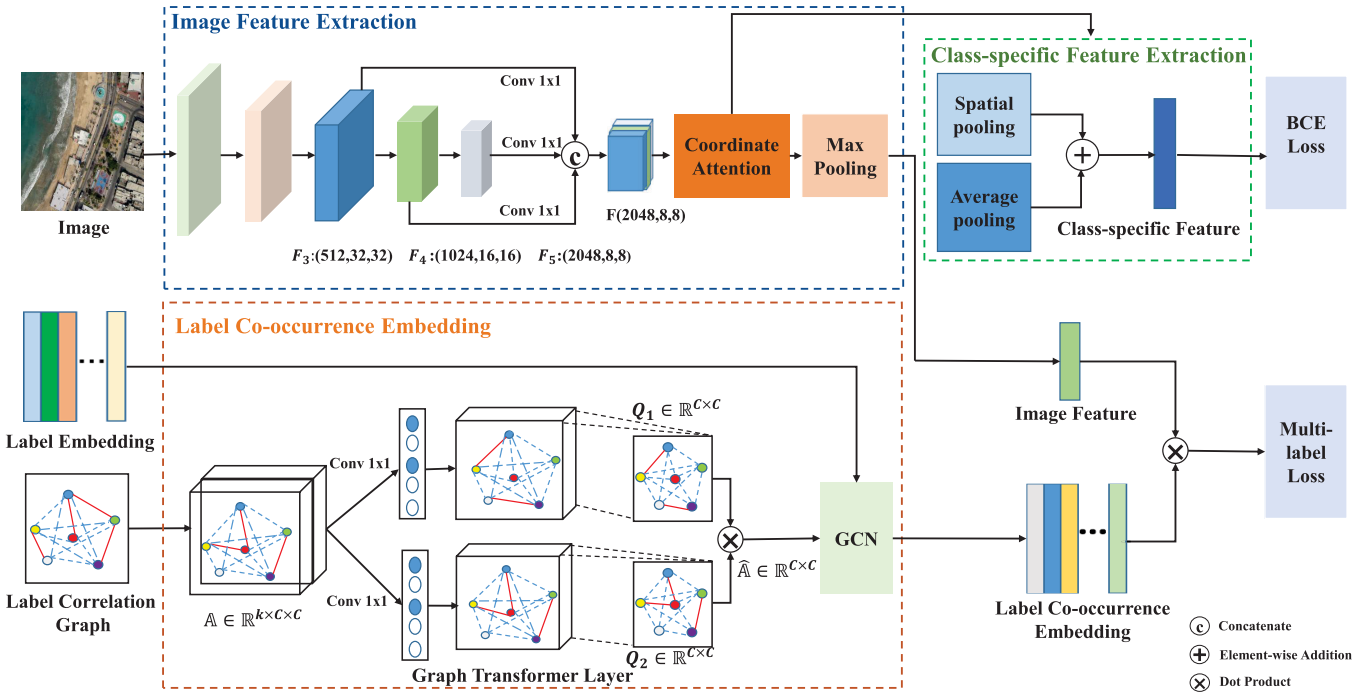


Fig. 2. Overall framework of our model for multilabel remote sensing image classification. It consists of the IFEM, the LCEM, and the CFEM.

nodes to generate reliable label co-occurrence embedding. In addition, we establish the class-specific feature extraction as an auxiliary task, which can improve the model's local and global feature extraction ability, to assist the multilabel image classification task. Finally, the output image features and label co-occurrence embedding are scored using the dot product to obtain the predicted labels for each image. In this way, the performance of the main multilabel remote sensing image classification task will be significantly improved with the help of the auxiliary task.

A. Image Feature Extraction

Remote sensing images often contain a large number of objects with different sizes. Although the deep features can represent the general semantic information of the image, with the increase of the convolutional layer, some smaller objects in the remote sensing image will be ignored, so the single-scale image feature cannot fully represent the whole information of a given remote sensing image. Shallow features have rich detail information and smaller receptive fields to identify smaller objects, while deep features have rich semantic information and larger receptive fields to identify larger objects. Therefore, there is a necessity to take the fusion of different scale features of remote sensing images into consideration.

To extract the fundamental image features, ResNet-50 is employed. Specifically, the image features of different convolutional layers can be expressed as

$$\{F_m\}_{m=1}^{5} = \text{ResNet}(I, \theta) \tag{1}$$

where I is the input remote sensing image, θ represents the parameters of ResNet, and F_m is the output feature map of each convolution layer.

Since F_1 and F_2, main contain low-level semantic features, we only perform feature fusion using F_3, F_4, and F_5. The features of these three scales are concatenated in the channel dimension to obtain the multiscale feature F. The above process can be formulated as

$$F = [(c_1(F_3), c_2(F_4), c_3(F_5))] \tag{2}$$

where c_1, c_2, and c_3 represent different 1×1 convolution kernels, and their output dimensions are 512, 512, and 1024, respectively. $[\cdot, \cdot, \cdot]$ denotes the concatenation operation in the channel dimension.

To enhance the representation ability of the fused feature, positional information also needs to be considered to generate spatially selective attention maps. The spatial structure of remote sensing images is very complex, so it is important to consider the positional information when extracting high-level semantic features of remote sensing images. Therefore, we introduce coordinate attention (CA), which embeds position information into channel attention to efficiently integrate spatial coordinate information [51]. Compared to the squeeze-and-excitation (SE) [52] and the convolutional block attention module (CBAM) [53], CA can capture long-term dependencies along one spatial direction, as well as preserve precise position information along another spatial direction.

Specifically, the global pooling is decomposed into horizontal pooling and vertical pooling along the input feature map to obtain the position information in two directions. Given the input F, the pooling in different directions of the cth channel can be derived as

$$F_c^h(h) = \frac{1}{W} \sum_{0\le i<W} F_c(h, i) \tag{3}$$

$$F_c^w(w) = \frac{1}{H} \sum_{0\le j<H} F_c(j, w). \tag{4}$$

The above two operations aggregate features along two spatial directions, respectively, which are used to generate

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the CA map. Specifically, we concatenate F^h and F^w with position information and use 1×1 convolution to compress the channel dimension. The process can be formulated as

$$u = \delta(Z([F^h, F^w])) \tag{5}$$

where Z is 1×1 convolution that converts the output feature map into $u \in \mathbb{R}^{C/r \times (H+W)}$ to reduce the model complexity, and δ denotes the sigmoid function. In order to get the CA map, u is split into two separate tensors, namely, $u^h \in \mathbb{R}^{C/r \times H}$ and $u^w \in \mathbb{R}^{C/r \times W}$, and 1×1 convolution is performed on them, respectively. The process can be formulated as

$$\mathbf{g}^h = \sigma(Z_h(u^h)) \tag{6}$$

$$\mathbf{g}^w = \sigma(Z_w(u^w)). \tag{7}$$

Finally, the g^h and g^w are used for coordinating the attention weights, respectively. f_{CA_c} is the output of the CA F_{CA} of the cth channel and can be written as

$$f_{\mathrm{CA}_c}(i, j) = f_c(i, j) \times g_c^h(i) \times g_c^w(j). \tag{8}$$

F_{CA} contains long-range dependencies in one spatial direction and preserves precise positional information in other spatial directions, which helps the model locate the objects of interest more accurately. To exploit the global contextual information, F_{CA} is squeezed into a channelwise descriptor F_G by using the global max pooling.

B. Label Co-Occurrence Embedding

In multilabel image classification tasks, label correlation can provide important complementary information. However, the label correlation matrix based on GCN overly relies on label node pairs with strong correlation in the graph while easily ignoring the positive effect of weakly correlated label node pairs, resulting in difficulties in accurately characterizing the association between multiple objects in complex scenarios. Graph transformer layer [54] can identify useful underlying correlations between unconnected nodes in a graph and generate a new graph structure to learn more efficient graph node representations. Inspired by this, we introduce the graph transformer layer into the label co-occurrence embedding process. In this way, the initial label correlation graph is updated by the graph transformer layer to improve the correlation of potential label node pairs in the graph. Then, GCN is used to mine high-order correlation between label nodes to generate more accurate label co-occurrence embedding for image label prediction.

First, we construct a label co-occurrence matrix. Specifically, the labels of each image are traversed to calculate the number of times the label pair appears and get the matrix $M \in \mathbb{R}^{C \times C}$. Concretely, C is the number of categories, and M_{ij} denotes the concurring times of label L_i and label L_j. The probability matrix can be calculated by

$$\mathrm{P}_{ij} = M_{ij}/N_i \tag{9}$$

where N_i represents the total number of labels for the ith category in the dataset, and all P_{ij} constitute a label co-occurrence matrix P.

In order to eliminate the interference of noise, a single cut-off threshold is usually applied in the normalization of the label co-occurrence matrix, which may lead to indistinguishable representations. We use multiple thresholds $t = [t_1, t_2, \ldots, t_K]$ to generate the label co-occurrence tensor $\mathbb{A} \in \mathbb{R}^{K \times C \times C}$ which consists of $\mathbb{A}_k \in \mathbb{R}^{C \times C}$, $k = [1, 2, \ldots, K]$. We set $\mathbb{A}_1 \in \mathbb{R}^{C \times C}$ as the identity matrix I and for $k \geq 2$ we have

$$\mathbb{A}_{kij} = \begin{cases} 1, & \text{if } \mathrm{P}_{ij} \in [t_{k-1}, t_k], \quad i \neq j \\ 0, & \text{otherwise.} \end{cases} \tag{10}$$

The graph transformer layer first selects graph structures Q_1 and Q_2 from the label co-occurrence tensor $\mathbb{A}$ and then learns a new graph structure through matrix multiplication of Q_1 and Q_2. Specifically, two softly chosen probability matrices $Q_1, Q_2 \in \mathbb{R}^{C \times C}$ are computed via two 1×1 convolutions as follows:

$$Q_1 = \psi\left(\mathbb{A}, \mathrm{softmax}\left(W_\psi^1\right)\right) \tag{11}$$

$$Q_2 = \psi\left(\mathbb{A}, \mathrm{softmax}\left(W_\psi^2\right)\right) \tag{12}$$

where ψ is the convolution layer, and $W_\psi^1, W_\psi^2 \in \mathbb{R}^{1 \times 1 \times K}$ are parameters to be learned.

The final transformed matrix $\hat{A} \in \mathbb{R}^{C \times C}$ is formulated as

$$\hat{A} = \eta(Q_1 Q_2 + I) \tag{13}$$

where I is the identity matrix, $\eta(A) = \mathcal{D}^{-1/2} A \mathcal{D}^{-1/2}$ is the Laplacian normalization.

In order to describe the LCEM more clearly, we use the figure to illustrate the detailed construction process of the graph transformer layer, as shown in Fig. 3.

After constructing the label co-occurrence matrix, we use GCN to extract the correlation between labels to generate reliable label co-occurrence embedding. GCN can propagate information between nodes to update node representation. Specifically, for the lth GCN layer, the input node feature representation is $H^l \in \mathbb{R}^{n \times d}$, and the output $H^{l+1} \in \mathbb{R}^{n \times d'}$ can be written as a nonlinear operation

$$H^{l+1} = g\left(\mathrm{A}\mathrm{H}^l W^l\right) \tag{14}$$

where $W^l \in \mathbb{R}^{d \times d'}$ denotes a learnable transformation matrix, $A \in \mathbb{R}^{C \times C}$ denotes the correlation matrix, and g is LeakyReLU function.

In this module, the matrix $\hat{A}$ is used as the correlation matrix between nodes in the graph, and the Glove word embedding vector of the label is used as the node embedding E. The label co-occurrence embedding $\mathcal{W} \in \mathbb{R}^{C \times D}$ can be formulated as

$$\mathcal{W} = \mathrm{GCN}(\hat{A}, E). \tag{15}$$

C. Class-Specific Feature Extraction

In remote sensing images, different objects have different positions and sizes, and simple pooling is not enough to learn the feature representation suitable for all objects. The CFEM learns class-specific spatial pooling coefficients for each class, then performs class-specific spatial pooling and average pooling operations on the fused features and adds them element-wise to generate class-specific features for each class.

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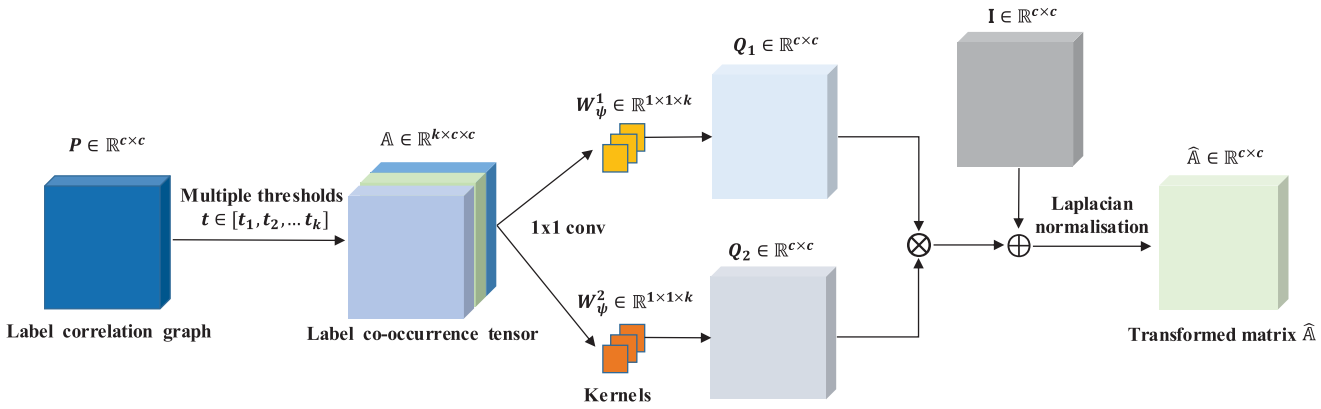


Fig. 3. Detailed construction process of the graph transformer layer.

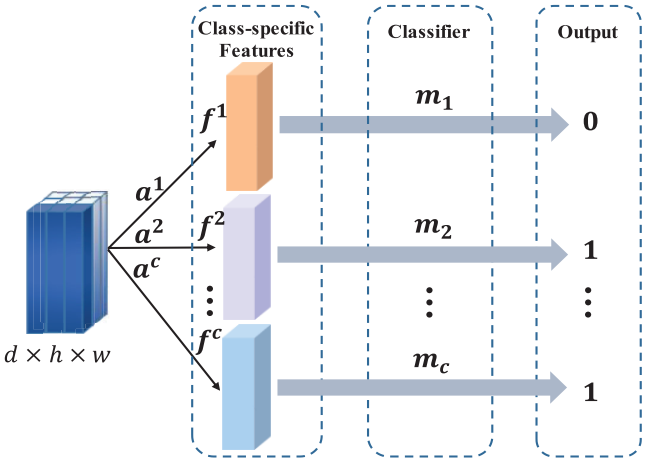


Fig. 4. Class-specific features of each class are used for the binary classification auxiliary task.

Meanwhile, the auxiliary task uses the same dataset as the main task and provides additional supervision during training, which can enhance the feature representation ability of the model to effectively improve the performance of the main task. As shown in Fig. 4, we establish a binary classification auxiliary task that uses the class-specific features to perform binary classification for each class to assist multilabel remote sensing image classification.

Specifically, the input $F_{\mathrm{CA}} \in \mathbb{R}^{d\times h\times w}$ can be converted to $F_{\mathrm{CA}}^1, F_{\mathrm{CA}}^2, \ldots, F_{\mathrm{CA}}^L$ ($F_{\mathrm{CA}}^i \in \mathbb{R}^d$, $L = h \times w$). We can define the class-specific attention scores for the ith class and jth location as

$$s_j^i = \frac{\exp\left(\left(F_{\mathrm{CA}}^j\right)^{\mathrm{T}} m_i\right)}{\sum_{k=1}^{L} \exp\left(\left(F_{\mathrm{CA}}^k\right)^{\mathrm{T}} m_i\right)} \tag{16}$$

where $m_i \in \mathbb{R}^d$ is a learnable parameter for the ith, and $\sum_{j=1}^{L} s_j^i = 1$.

The spatial features for ith class can be defined as the following equation:

$$a^i = \sum_{k=1}^{L} s_k^i F_{\mathrm{CA}}^k \tag{17}$$

where s_k^i is the attention scores for the ith class and can be viewed as the probability that class i appears at position k.

We can obtain the class-specific features f^i for the ith class

$$f^i = g + \eta a^i \tag{18}$$

where g is the average pooling features for F_{CA}, and $\eta \in [0, 1]$ is a weighted factor to balance these two features. Finally, each class-specific features are sent to a classifier for a binary classification auxiliary task to obtain the final logits

$$\hat{y}_B = \left(m_1^{\mathrm{T}} f^1, m_2^{\mathrm{T}} f^2, \ldots, m_C^{\mathrm{T}} f^C\right) \tag{19}$$

where m_i is the binary classifier for the ith class, and C is the number of the class.

D. Loss Function

The output image features F_G and label co-occurrence embedding $\mathcal{W}$ are scored to obtain the predicted label for each image

$$\hat{y}_m = \mathcal{W} F_G. \tag{20}$$

We use multilabel classification loss as the loss function, calculated as

$$\mathcal{L}_m = \sum_{c=1}^{C} y^c \log\left(\sigma\left(\hat{y}_m^c\right)\right) + \left(1 - y^c\right) \log\left(1 - \sigma\left(\hat{y}_m^c\right)\right) \tag{21}$$

where $\sigma(\cdot)$ is the sigmoid function.

At the same time, we use the BCE loss function as the loss function for the binary classification task of each class-specific feature, which is calculated as follows:

$$\mathcal{L}_b = -\sum_{i=1}^{n} \sum_{j=1}^{c} y_{ij} \log \hat{y}_{bij} + (1 - y_{ij}) \log\left(1 - \hat{y}_{bij}\right). \tag{22}$$

Finally, the above $\mathcal{L}_b$ and $\mathcal{L}_m$ are integrated to train the whole network in an end-to-end manner with the following total loss:

$$\mathcal{L} = \mathcal{L}_b + \lambda \mathcal{L}_m \tag{23}$$

where $\lambda \in [0, 1]$ is a weighted factor to balance these two loss functions.

In order to make it clear, we summarize the process of the proposed method in Algorithm 1.

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Algorithm 1 Proposed Method

Input: Image I, label embedding E, label co-occurrence matrix P, and the multi-label y.

Output: Predicted multi-label $\hat{y}_m$.

1: Initialize the parameters θ of ResNet, the parameters W of GCN, the parameters W_ψ of the graph transformer layer, and the parameters m of the binary classifier;
2: **for** the epoch $= 0$ to the max epoch **do**
3: For image I, extract visual features F_G with the image feature extraction module;
4: For label embedding E, label co-occurrence matrix P, obtain label co-occurrence embedding $\mathcal{W}$ with the label co-occurrence embedding module;
5: Generate class-specific features $f^i \leftarrow g + \eta a^i$, and perform binary classification for i-th class;
6: Obtain the predicted label $\hat{y}_m \leftarrow \mathcal{W} F_G$;
7: Back-propagate the gradients
8: Update network parameters θ, W, W_ψ, m by minimizing $\mathcal{L} \leftarrow \mathcal{L}_b + \lambda \mathcal{L}_m$;
9: **end for**

IV. Experimental Results

In this section, we conduct several experiments to verify the effectiveness of our proposed method with seven subsections, including commonly-used datasets, evaluation metrics, implementation details, comparison with other state-of-the-art methods, ablation study, per-class case studies, comparison of model efficiency, visualization results, and failure case studies, respectively.

A. Datasets

1) UCM Multilabel Dataset: The UCM multilabel dataset [55] is recalibrated from the UCM dataset, which is produced by the United States Geological Survey (USGS) National Map and contains 2100 images with a size of 256 × 256 × 3 and a total of 21 scene categories. The UCM multilabel dataset reassigns multiple object-level labels to each image based on the objects presented in the image. There are a total of 17 object-level labels including airplanes, bare-soil, buildings, cars, chaparral, court, dock, field, grass, mobile-home pavement, sand, sea, ship, tanks, trees, and water. Fig. 5(a) gives the specific number of each class label, and Fig. 6 illustrates some multilabel examples from this dataset.

2) AID Multilabel Dataset: The AID multilabel dataset [56] is selected from the AID dataset, which is produced by Wuhan University and contains 10 000 images with a size of 600 × 600 × 3 and covering a total of 30 scene categories. The AID multilabel dataset selects 100 images of each scene category from the AID dataset, including 3000 multilabel images, which are consistent with the object-level labels of the UCM dataset. Fig. 5(b) gives the specific number of each class label, and Fig. 7 illustrates some multilabel examples from this dataset.

3) DFC15 Multilabel Dataset: The DFC15 multilabel dataset [56] is recreated from the DFC15 dataset. It contains 3342 images with a size of 256 × 256 × 3 and a total of eight object-level labels including building, boat, car, clutter, impervious, water, vegetation, and trees. Fig. 5(c) gives the specific number of each class label, and Fig. 8 illustrates some multilabel examples from this dataset.

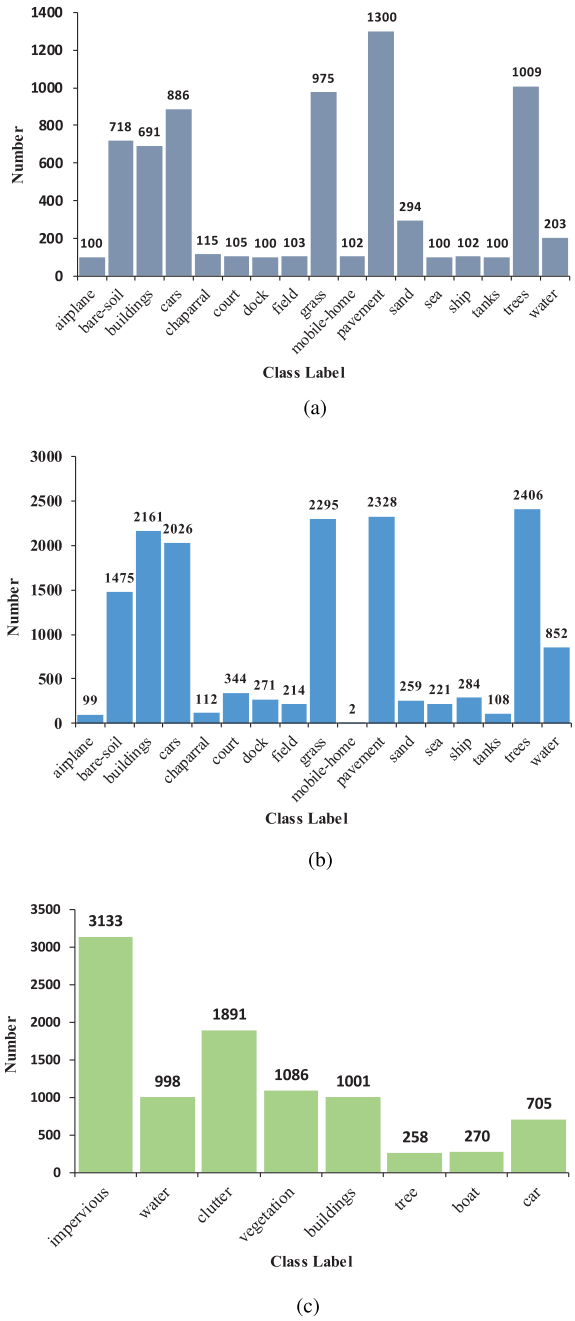


Fig. 5. Label statistics for (a) UCM, (b) AID, and (c) DFC15 multilabel datasets.

Images in the above three multilabel remote sensing image datasets contain different objects with different sizes and have strong interclass similarities, which makes it difficult for the model to accurately predict the category of each label. Meanwhile, the number of each class in the above three multilabel remote sensing image datasets varies greatly, which leads to the data imbalance problem affecting the model performance.

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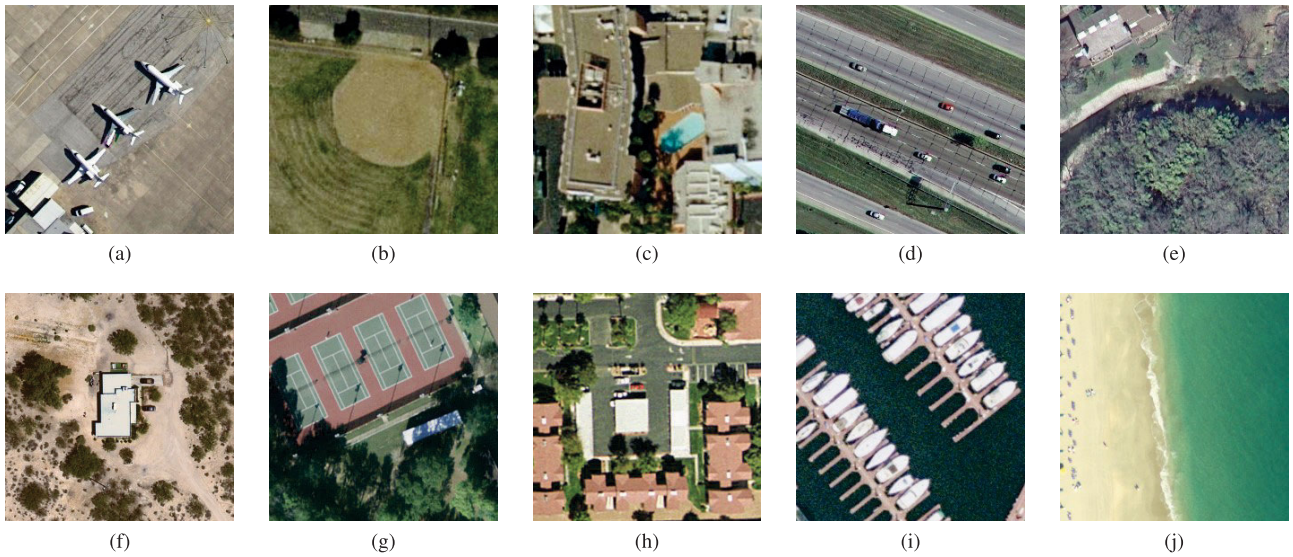


Fig. 6. Samples of the UCM multilabel dataset from different scenes. (a) Airplanes, cars, and pavement. (b) Bare-soil, grass, pavement, and trees. (c) Buildings and trees. (d) Cars, grass, and pavement. (e) Bare-soil, buildings, grass, sand, trees, and water. (f) Buildings, cars, chaparral, sand, and trees. (g) Court, grass, and trees. (h) Buildings, cars, grass, pavement, and trees. (i) Dock, ship, and water. (j) Sand and sea.

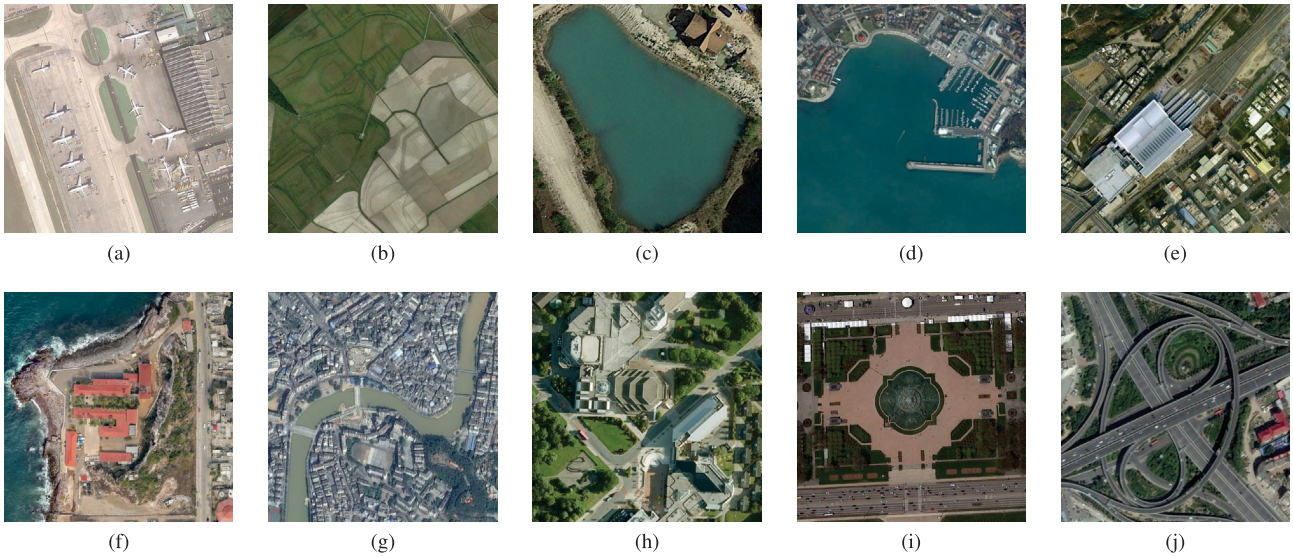


Fig. 7. Samples of the AID multilabel dataset from different scenes. (a) Airplanes, buildings, grass, cars, pavement, and trees. (b) Bare-soil, buildings, fields, and trees. (c) Bare-soil, buildings, cars, chaparral, grass, trees, and water. (d) Buildings, cars, docks, grass, pavement, sea, ship, and trees. (e) Bare-soil, buildings, cars, grass, pavement, and trees. (f) Bare-soil, buildings, cars, grass, pavement, sea, ship, and trees. (g) Bare-soil, buildings, cars, docks, grass, pavement, ships, trees, and water. (h) Buildings, cars, grass, pavement, and trees. (i) Buildings, cars, grass, pavement, and trees. (j) Bare-soil, buildings, cars, grass, pavement, and trees.

B. Evaluation Metrics

To comprehensively verify the performance of our proposed method, we use example-based and label-based metrics for evaluating the classification performance. Specifically, the example-based metrics $F1$ score, $F2$ score, precision, and recall are used to evaluate the performance of different models in general, and the label-based metrics $F1$ score, precision, and recall are used to evaluate the classification performance of the network for each category.

1) Example-Based Metrics: For this kind of metrics, precision P_e and recall R_e are defined as follows:

$$\mathrm{P}_e = \frac{1}{n}\sum_{i=1}^{n}\frac{\mathrm{TP}_e}{\mathrm{TP}_e + \mathrm{FP}_e} \tag{24}$$

$$\mathrm{R}_e = \frac{1}{n}\sum_{i=1}^{n}\frac{\mathrm{TP}_e}{\mathrm{TP}_e + \mathrm{FN}_e} \tag{25}$$

where TP_e represents the number of positive labels predicted to be positive in the sample, FP_e represents the number of positive labels predicted to be negative in the sample, and FN_e represents the number of negative labels predicted to be positive in the sample. n is the total number of the images.

Then, the $F1$ score and $F2$ score can be calculated from the precision and recall values

$$F\beta_e = \left(1+\beta^2\right)\frac{\mathrm{P}_e R_e}{\beta^2 P_e + \mathrm{R}_e}, \quad \beta = 1, 2. \tag{26}$$

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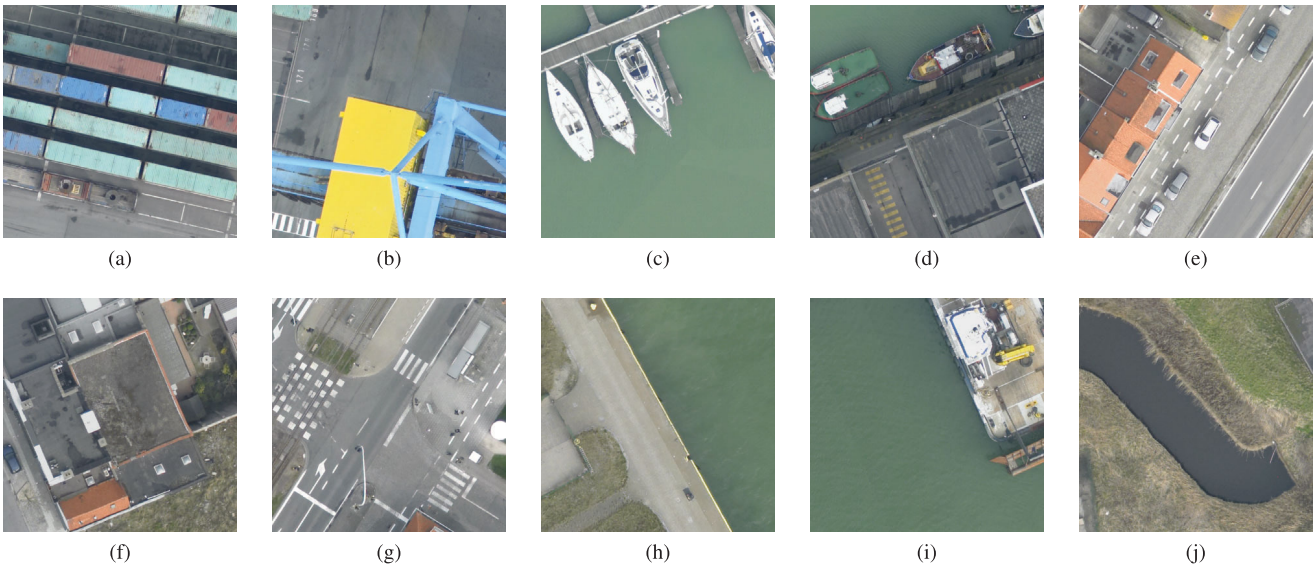


Fig. 8. Samples of the DFC15 multilabel dataset from different scenes. (a) Impervious and cluttered. (b) Impervious and cluttered. (c) Water, clutter, and boat. (d) Impervious, water, buildings, and boats. (e) Impervious, clutter, buildings, and cars. (f) Impervious, vegetation, and buildings. (g) Impervious, clutter, tree, and car. (h) Impervious, water, and vegetation. (i) Impervious, water, and boat. (j) Impervious, water, clutter, and vegetation.

2) Label-Based Metrics: The label-based metrics precision P_l, and recall R_l are defined as

$$\mathrm{P}_l = \frac{1}{C}\sum_{j=1}^{C}\frac{\mathrm{TP}_l}{\mathrm{TP}_l + \mathrm{FP}_l} \tag{27}$$

$$\mathrm{R}_l = \frac{1}{C}\sum_{j=1}^{C}\frac{\mathrm{TP}_l}{\mathrm{TP}_l + \mathrm{FN}_l} \tag{28}$$

where TP_l represents the number of positive labels predicted to be positive in a class, FP_l represents the number of positive labels predicted to be negative in a class, and FN_l represents the number of negative labels predicted to be positive in a class. C is the number of the classes.

Similarly, the $F1$ score and $F2$ score can be calculated from the precision and recall values

$$F\beta_l = \left(1+\beta^2\right)\frac{\mathrm{P}_l R_l}{\beta^2 P_l + \mathrm{R}_l}, \quad \beta = 1, 2. \tag{29}$$

C. Implementation Details

In our experiments, we use the pretrained ResNet-50 to extract image features. F_3–F_5 are the feature maps of conv3–conv5 in ResNet-50, respectively, and F is the multiscale feature. The sizes of F_3, F_4, F_5, and F are $512 \times 32 \times 32$, $1024 \times 16 \times 16$, $2048 \times 8 \times 8$, and $2048 \times 8 \times 8$, respectively. For the graph transformer layer, without otherwise stated we set multiple thresholds $t = [0.2, 0.4, 1.0]$ and GCN consists of two layers with output dimensionality of 1024 and 2048. We set feature balance factor $\eta = 0.2$ for the class-specific feature extraction and loss balance factor $\lambda = 0.5$ for the loss function. For the label embedding, we adopt a 300-dim Glove [57] word embedding vector trained on the Wikipedia dataset. During training, the input images are randomly cropped and resized into 256×256 with random horizontal flips for data augmentation. We randomly select 70% of the samples from each of the three datasets as a training set, 10% as a validation set, and 20% as a test set. The model is trained with the Adam [58] optimizer and the learning rate is initially set to 0.001, which decays every 20 epochs with a factor of 0.1. The batch size is set to 32, and the model is trained for 60 epochs in total. The model is implemented on PyTorch 1.12.1 and trained on NVIDIA GeForce GTX 3090 GPU.

D. Comparisons With State-of-the-Arts

For a comprehensive evaluation, we compare the proposed method with the following classical state-of-the-art multilabel classification methods.

1) GRN [59]: This method can use the graph structure fed back into network learning to model the relationship between different scenes.

2) MSGM [60]: This method extracts image features through spatial pyramid convolution and uses GCN to mine the correlation between labels.

3) ML-CG [29]: This method completes the label information by adding ConceptNet to generate a concept graph and uses GCN to extract label correlation information.

4) AL-RN [61]: This method extracts high-level label features and uses an attention region extraction module to localize regions within these features to generate attention label-specific features.

5) CA-BiLSTM [2]: This method extracts fine-grained semantic feature maps of images and captures class-specific features with an attention layer, and finally models class dependencies based on a bidirectional LSTM network.

6) ML-GCN [26]: This method uses the pretrained ResNet for image feature extraction and GCN for label correlation extraction.

7) SR-Net [30]: This method localizes semantic attention regions in features extracted by deep CNNs and generates discriminative category representations.

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TABLE I
PERFORMANCE COMPARISONS WITH STATE-OF-THE-ART METHODS ON THREE MULTILABEL REMOTE SENSING DATASETS (%)

Method	UCM				AID				DFC15			
	F1	F2	P	R	F1	F2	P	R	F1	F2	P	R
GRN	92.40	92.66	91.98	92.83	91.93	91.42	92.79	91.08	96.24	96.07	96.53	95.95
MSGM	84.66	85.15	83.86	85.48	-	-	-	-	93.65	93.08	94.61	92.71
ML-CG	85.42	88.08	81.34	89.94	-	-	-	-	95.88	96.01	95.68	96.09
AL-RN	87.93	87.41	88.81	87.07	89.96	89.35	91.00	88.95	-	-	-	-
CA-BilSTM	83.11	86.56	77.94	89.02	89.01	89.00	89.03	88.99	-	-	-	-
ML-GCN	91.43	91.07	92.03	90.83	90.92	90.12	92.28	89.60	95.25	94.61	96.33	94.20
SR-Net	88.67	89.11	87.96	89.40	89.97	90.30	89.42	90.52	-	-	-	-
SIGNA	93.96	93.94	94.00	93.93	92.96	92.66	93.46	92.46	97.46	97.07	**98.12**	96.81
Auxiliary Task	93.91	93.75	**94.17**	93.65	92.71	92.30	93.42	92.03	97.30	97.01	97.80	96.81
Ours	**94.26**	**94.39**	94.06	**94.47**	**93.47**	**93.08**	**94.13**	**92.82**	**97.68**	**97.47**	98.02	**97.34**

8) SIGNA [34]: This method guides image features into a semantic feature space embedded with label information, and extracted visual features through global channel attention.

For a fair comparison, all compared methods select ResNet-50 as the backbone. The baseline is ML-GCN which uses the pretrained ResNet-50 for image feature extraction and GCN for label correlation extraction. Our experiments are performed on three multilabel remote sensing datasets, namely, UCM, AID, and DFC15. $F1$, $F2$, P, and R stand for $F1$ score, $F2$ score, precision and recall, respectively. As shown in Table I, higher values indicate superior classification performance. The optimal results are shown in bold.

On the UCM multilabel dataset, our method achieves the best performance in terms of all studied metrics compared to other methods. Specifically, our proposed methods achieve the $F1$ score of 94.26%, the $F2$ score of 94.39%, the P of 94.06%, and the R of 94.47%, respectively, which are 0.30%, 0.45%, 0.06%, and 0.54% higher than the SIGNA model.

On the AID multilabel dataset, our method achieves the best performance in terms of all studied metrics compared to other methods. Specifically, our proposed methods achieve the $F1$ score of 93.47%, the $F2$ score of 93.08%, the P of 94.13%, and the R of 92.82%, respectively, which are 0.51%, 0.42%, 0.67%, and 0.36% higher than the SIGNA model. Although the AID multilabel dataset and the UCM multilabel dataset have the same object-level labels, the AID multilabel dataset generally contains more objects per image than the UCM multilabel dataset. Therefore, most methods perform lower on the AID multilabel dataset than the UCM multilabel dataset, which shows that the multilabel image classification task is more challenging on the AID multilabel dataset than on the UCM multilabel dataset.

On the DFC15 multilabel dataset, the proposed method still outperforms all the other methods in most of the metrics except the precision. Our proposed methods achieve the $F1$ score of 97.68%, the $F2$ score of 97.47%, and the R of 97.34%, respectively, which are 0.22%, 0.40%, and 0.53% higher than the SIGNA model.

It should be noted that although both our method and SIGNA introduce the visual attention mechanism, our method can more effectively identify objects of different sizes and distinguish similar objects through the multiscale fusion module and the CFEM. Compared with ML-GCN, which also uses GCN for label relation feature extraction, the performance of our method far exceeds those of ML-GCN on all three datasets. This observation demonstrates that our method can mine more fine-grained features with the aid of the auxiliary task, which assists in making more accurate feature matching between the remote sensing image and multiple labels.

To further verify the advantages of the multitask framework, we give the multilabel classification performance comparison by only using the binary classification auxiliary task with other existing methods. As shown in the line marked as "auxiliary task" in Table I, the classification accuracy of only using the binary classification auxiliary task is close to that of the existing state-of-the-art methods. Therefore, this confirms that the multitask framework can efficiently learn and integrate the underlying semantic features of remote sensing images between two related tasks (i.e., the binary classification task and the multilabel image classification task), which effectively compensates for the limitations of single-task learning in the face of limited data to significantly improve the generalization performance of the model, thus enabling the proposed method to outperform the existing state-of-the-art methods.

E. Ablation Study

The proposed multilabel remote sensing image classification framework is composed of three modules, including the IFEM, the LCEM, and the CFEM. To evaluate the effectiveness of these modules, we compare our method with the following methods.

1) ML-GCN: ML-GCN uses the pretrained ResNet-50 for image feature extraction and GCN for label correlation extraction.

2) ML-GCN + Class-Specific Feature Extraction: CFEM extracts class-specific features for the binary classification auxiliary task to assist the main multilabel image classification task.

3) ML-GCN + Graph Transformer Layer: The graph transformer layer updates the graph structure to improve the correlation of potentially useful label node pairs in the graph.

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TABLE II
MODULE EFFECTIVENESS EVALUATION OF THE PROPOSED METHOD ON THE UCM MULTILABEL DATASET (%)

Models	F1	F2	P	R
ML-GCN	91.43	91.07	92.03	90.83
ML-GCN+Class-specific Feature Extraction	92.67	92.30	93.30	92.05
ML-GCN+Graph Transformer Layer	92.93	92.41	93.82	92.06
ML-GCN+Graph Transformer Layer+ Class-specific Feature Extraction	94.08	93.80	**94.55**	93.62
ML-GCN+Graph Transformer Layer+ Class-specific Feature Extraction+ Multi-scale Fusion Strategy and Visual Attention Mechanism (Proposed Method)	**94.26**	**94.39**	94.06	**94.47**

4) ML-GCN + Graph Transformer Layer + Class-Specific Feature Extraction: Graph transformer layer and CFEM are used simultaneously for multilabel image classification tasks.

We verify the effectiveness of each module of our framework on the UCM multilabel dataset, and the results are shown in Table II. In the ML-GCN method, we use a single threshold t to generate the label co-occurrence matrix $A \in \mathbb{R}^{C\times C}$. During the training process, the label co-occurrence tensor A is fixed as the input of GCN. As a result, the label co-occurrence matrix based on GCN overly relies on label node pairs with strong correlation in the graph while easily ignoring the positive effect of weakly correlated label node pairs. In ML-GCN + graph transformer layer method, we use multiple thresholds $t = [t_1, t_2, \ldots, t_K]$ to generate the label co-occurrence tensor $\mathbb{A} \in \mathbb{R}^{K\times C\times C}$. We update the label correlation graph structure using the graph transformer layer to accurately identify label node pairs with potential correlation. From the comparison of the ML-GCN + graph transformer layer method and the ML-GCN method, the values of $F1$ score, $F2$ score, P, and R have increased by 1.50%, 1.34%, 1.79%, and 1.23%, respectively. Therefore, we can conclude that the graph transformer layer can improve the impact of weak correlation on model performance for image label prediction to perform more efficient graph convolutions on labels.

The effectiveness of the binary classification auxiliary task is evaluated by comparing the ML-GCN method with the ML-GCN + class-specific feature extraction method. The values of $F1$ score, $F2$ score, P, and R have increased by 1.24%, 1.23%, 1.27%, and 1.22%, respectively. We can conclude that the CFEM improves the model's local and global feature extraction ability to largely improve the classification accuracy. The comparison of the ML-GCN + graph transformer layer method with the ML-GCN + graph transformer layer + class-specific feature extraction method further indicates the effectiveness of integrating the CFEM. From the comparison of the ML-GCN + graph transformer layer + class-specific feature extraction method and the proposed method, the values of $F1$ score, $F2$ score, and R have increased by 0.18%, 0.59%, and 0.85%, respectively. It can be inferred that the proposed method fuses multilevel features of remote sensing images and enhances the role of dominant features for the classification task, thereby outputting multiple object-level semantic labels that are closely related to the content of the remote sensing image. The proposed method generally outperforms the other three methods, which indicates the effectiveness of integrating these modules.

TABLE III
COMPARISON OF THE MODEL WITH SINGLE THRESHOLD AND WITH MULTIPLE THRESHOLD t (%)

t	F1	F2	P	R
0.2	91.43	91.07	92.03	90.83
[0.2,0.4,1]	**92.93**	**92.41**	**93.82**	**92.06**
[0.2,0.6,1]	92.27	91.87	92.94	91.61
[0.2,0.8,1]	91.67	91.39	92.14	91.20

TABLE IV
$F1$, P, AND R SCORES OF OUR PROPOSED METHOD AND BASELINE ON EACH CATEGORY OF THE UCM MULTILABEL DATASET (%)

Labels	ours			baseline		
	F1	P	R	F1	P	R
airplane	**100.00**	**100.00**	**100.00**	100.00	100.00	100.00
bare-soil	**84.42**	**87.14**	**81.88**	76.77	73.94	79.86
buildings	**93.53**	**93.89**	**93.18**	88.97	86.43	91.67
cars	**91.95**	**92.49**	**91.43**	86.44	85.47	87.43
chaparral	**97.67**	**100.00**	**95.45**	97.67	100.00	95.45
court	**97.56**	**100.00**	**95.24**	95.00	100.00	90.48
dock	**100.00**	**100.00**	**100.00**	100.00	100.00	100.00
field	97.43	95.00	**100.00**	**100.00**	**100.00**	100.00
grass	**89.73**	**88.30**	**91.21**	86.63	84.38	89.01
mobile-home	97.56	95.24	**100.00**	**100.00**	**100.00**	100.00
pavement	**93.08**	**92.03**	94.16	92.59	88.34	**97.27**
sand	**93.10**	**91.53**	**94.74**	88.29	90.74	85.96
sea	**100.00**	**100.00**	**100.00**	100.00	100.00	100.00
ship	**100.00**	**100.00**	**100.00**	97.56	100.00	95.23
tanks	**97.43**	**100.00**	**95.00**	88.89	100.00	80.00
trees	**93.37**	**92.42**	**94.33**	88.89	89.12	88.66
water	**98.76**	97.56	**100.00**	96.10	**100.00**	92.50

Meanwhile, we analyze how model performance changes with different numbers of thresholds t. In the ML-GCN model, we use a single threshold t to generate the label co-occurrence matrix $A \in \mathbb{R}^{C\times C}$. In ML-GCN + Graph Transformer Layer model, we use multiple thresholds $t = [t_1, t_2, \ldots, t_K]$ to generate the label co-occurrence tensor $\mathbb{A} \in \mathbb{R}^{K\times C\times C}$. As shown in Table III, the model with multiple thresholds $t = [t_1, t_2, t_3]$ outperforms the model with a single threshold t in terms of all studied metrics. Specifically, the model with multiple thresholds $t = [0.2, 0.4, 1]$ achieves the $F1$ score of 92.93%, the $F2$ score of 92.41%, the P of 93.82%, and the R of 92.06%, respectively, which are 1.5%, 1.34%, 1.79%, and 1.23% higher than the model with a single threshold $t = 0.2$. In short, the model with multiple thresholds outperforms the model with a single threshold in terms of all studied metrics. However, an excessive number of thresholds can increase the computational complexity of the graph model. To make a good balance between accuracy and computational complexity, we choose the model with multiple thresholds $t = [0.2, 0.4, 1]$.

F. Per-Class Case Studies

As shown in Tables IV–VI, the per-class classification performance of our proposed method and baseline for each category on UCM, AID, and DFC15 multilabel datasets are represented by $F1$, P, and R scores, respectively. The optimal results in each category are shown in bold.

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TABLE V

$F1$, P, AND R SCORES OF OUR PROPOSED METHOD AND BASELINE ON EACH CATEGORY OF THE AID MULTILABEL DATASET (%)

Labels	ours			baseline		
	F1	P	R	F1	P	R
airplane	**94.74**	**94.74**	**94.74**	70.00	66.67	73.68
bare-soil	**86.40**	**89.96**	**83.11**	80.33	80.20	80.49
buildings	**96.30**	**95.41**	**97.20**	94.52	94.20	94.86
cars	**95.23**	**95.11**	**95.34**	93.27	94.92	91.67
chaparral	43.78	58.33	**35.00**	**45.16**	**63.64**	35.00
court	**70.80**	**86.96**	**59.70**	58.12	68.00	50.75
dock	78.26	83.72	73.47	**89.41**	**87.50**	**91.42**
field	**83.58**	**93.33**	**75.68**	77.33	76.31	78.37
grass	**96.32**	**95.09**	**97.59**	94.32	93.91	94.73
mobile-home	**0.00**	**0.00**	**0.00**	0.00	0.00	0.00
pavement	**98.28**	98.28	**98.28**	96.83	**98.44**	95.26
sand	**93.07**	**97.92**	**88.68**	85.47	78.13	94.33
sea	**97.67**	**97.67**	**97.67**	92.86	95.12	90.69
ship	73.91	79.07	**69.39**	**74.15**	**82.50**	67.34
tanks	**97.56**	**96.15**	**95.24**	95.23	95.23	95.23
trees	**96.30**	**94.75**	**97.91**	95.52	93.24	97.91
water	**81.78**	**85.71**	**77.65**	76.60	76.16	77.06

TABLE VI

$F1$, P, AND R SCORES OF OUR PROPOSED METHOD AND BASELINE ON EACH CATEGORY OF THE DFC15 MULTILABEL DATASET (%)

Labels	ours			baseline		
	F1	P	R	F1	P	R
impervious	**99.33**	**99.17**	**99.50**	97.70	96.75	98.68
water	**98.15**	**98.61**	**97.70**	95.35	96.70	94.04
clutter	**96.47**	97.11	**95.84**	94.18	**98.30**	90.39
vegetation	**97.96**	**97.75**	**98.20**	96.66	95.18	98.20
buildlings	**93.04**	**98.98**	**90.27**	90.81	93.67	88.11
trees	**91.84**	**91.84**	**91.84**	71.58	73.91	69.39
boat	**94.55**	**96.30**	**92.86**	84.00	85.45	75.00
car	**93.58**	**95.38**	**91.85**	91.19	94.44	88.15

Table IV shows the performance of different classes of the UCM multilabel dataset, and the result of our proposed method outperforms the baseline method in most classes. The $F1$ score, P, and R of our proposed method all reached 100% one the classes of airplane, dock, sea, and ship. Compared with the baseline method, the $F1$ scores of the proposed method are effectively improved on the classes of bare-soil, buildings, cars, courts, grass, pavement, sand, ships, tanks, trees, and water. Especially, the $F1$ scores of the proposed method demonstrate the highest improvement in the classes of bare-soil, sand, and tanks, by 7.65%, 4.81%, and 8.54%, respectively.

Table V shows the performance of different classes of the AID multilabel dataset, and the result of our proposed method outperforms the baseline method in most classes. In particular, the $F1$ scores of the proposed method on the classes of airplane, court, field, sand, and water are significantly ahead of the baseline method, with improvements of 24.74%, 12.68%, 6.25%, 7.6%, and 5.18%, respectively.

Table VI shows the performance of different classes of the DFC15 multilabel dataset, and the result of our proposed method outperforms the baseline in all classes except the clutter class. Compared to the baseline method, the $F1$ score of our proposed method has improved in each class. Especially, the per-class classification performance of trees and boats shows the greatest improvement in $F1$ scores compared to the baseline method, with the improvement of 20.26% and 10.55%, respectively.

TABLE VII

COMPARISONS OF MODEL EFFICIENCY

Model	FLOPs (M)	Params (M)	FPS	Inference time (ms)
ResNet-50	5398.55	25.56	417.27	2.40
ML-GCN	5396.37	25.91	476.44	2.10
SIGNA	5399.68	29.43	217.98	4.59
Ours	5960.04	29.62	337.33	2.96

G. Comparison of Model Efficiency

To further evaluate the performance of the proposed method, we study the model efficiency. Specifically, the computational complexity (measured by FLOPs), parameters, and time complexity (inference time and FPS) are investigated. Since most comparison methods do not publish these metrics, we only have made a data comparison with some methods. As reported in Table VII, the standard backbone (the original ResNet-50), the baseline method (ML-GCN), and the state-of-the-art method (SIGNA) are investigated for comparison. The FLOPs of our model are 5960.04M, which is slightly higher than that of ResNet-50, ML-GCN, and SIGNA. It can be observed that the number of parameters of our model is only 4.06M higher than that of ResNet-50, 3.71M higher than that of ML-GCN, and 0.19M higher than that of SIGNA. In terms of inference time, our model takes 2.96 ms inference time, slightly higher than that of ResNet-50 and ML-GCN and obviously lower than that of SIGNA. The comparison of model efficiency shows that the proposed model does not require much more resources compared with other models and makes a good balance between accuracy and efficiency for multilabel remote sensing image classification tasks.

H. Visualization

1) Example Predictions: To qualitatively compare our method with the baseline method, we randomly exhibit some example prediction results from of the UCM multilabel datasets, as shown in Table VIII. Incorrect predictions are shown in red. Compared with the baseline method, our proposed method can predict multiple object labels more accurately. In the first image of Table VIII, there are up to six real ground-truth labels including bare-soil, buildings, cars, pavement, trees, and water. In the second image of Table VIII, there are five real ground-truth labels including bare-soil, buildings, cars, pavement, and trees. And in the third image of Table VIII, there are four real ground-truth labels including buildings, grass, pavement, and trees. It can be noted that although the above object labels have different sizes and strong interclass similarities, our proposed method can identify them accurately.

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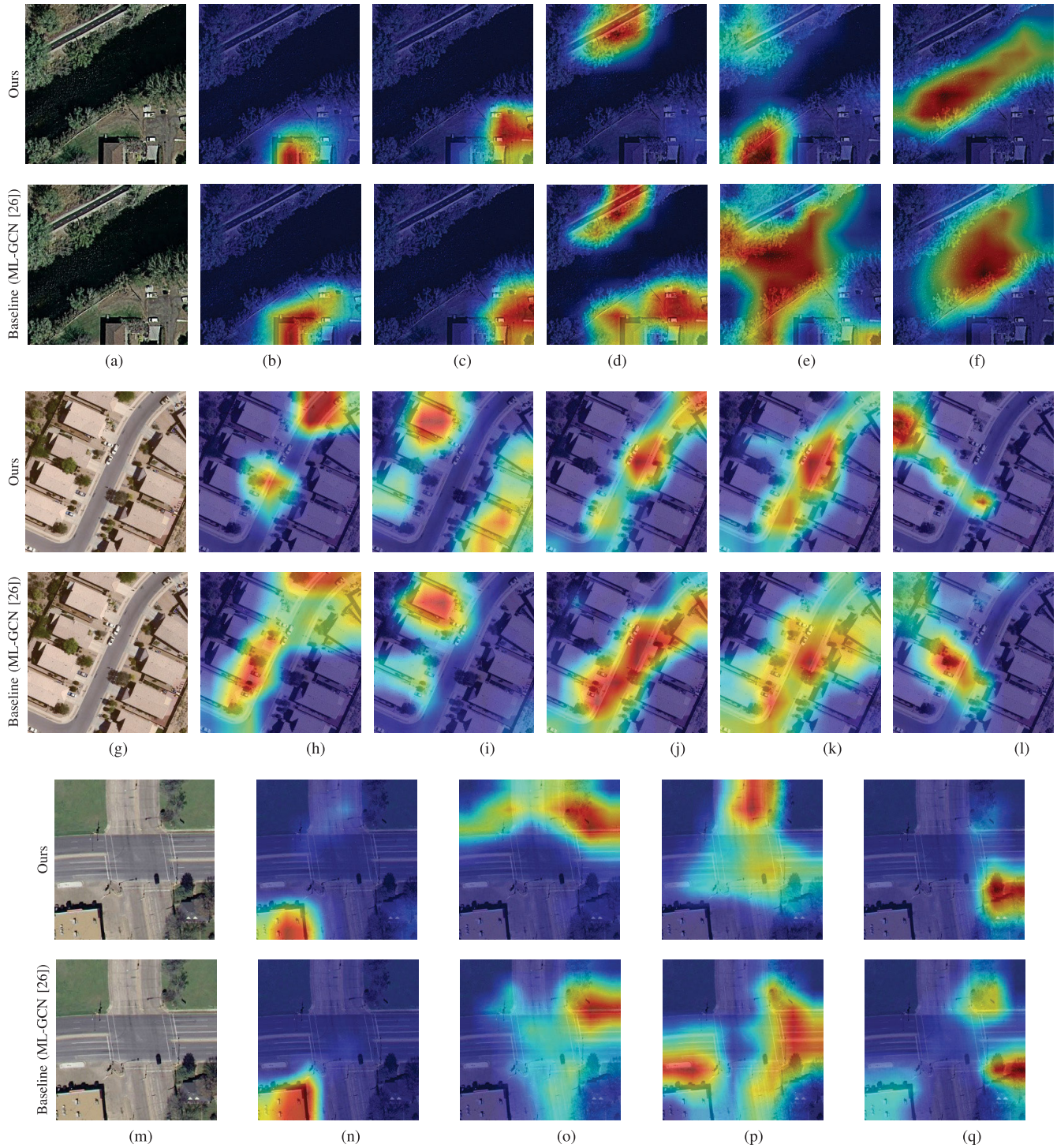
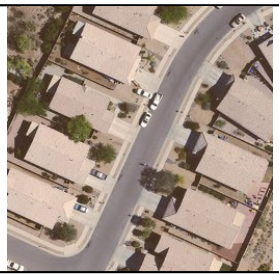


Fig. 9. Visualization results of three sample images from the UCM multilabel dataset. Regions with label-related semantic information are highlighted in the feature maps. (a) River. (b), (i), and (n) Buildings. (c) and (j) Cars. (d), (k), and (p) Pavement. (e), (l), and (q) Trees. (f) Water. (g) Dense residential. (h) Bare-soil. (m) Intersection. (o) Grass.

2) Feature Visualization: To further verify the effectiveness of our proposed method, we present a visualization analysis on three sample images of the UCM multilabel dataset. As shown in Fig. 9, regions with label-related semantic information are highlighted in the feature maps. The darker color region corresponds to higher attention. Compared with the baseline method, our proposed method can mine more fine-grained features to accurately identify objects of different sizes and distinguish similar objects in remote sensing images. For example, the scene-level label in Fig. 9(a) is the river, which contains objects of different sizes such as buildings, pavement, and water. Although the sizes of these objects are in a wide range, our proposed method can localize them accurately. Meanwhile, similar objects, such as grass and trees, can also be distinguished by using the proposed method.

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TABLE VIII
EXAMPLE PREDICTIONS ON THE UCM MULTILABEL DATASET

Image samples			
Ground Truth	bare-soil, buildings, cars, pavement, trees, water	bare-soil, buildings, cars, pavement, trees	buildings, grass, pavement, trees
Baseline (ML-GCN [26])	bare-soil, buildings, cars, pavement, trees, water	bare-soil, buildings, cars, pavement, trees	buildings, grass, pavement, trees
Ours	bare-soil, buildings, cars, pavement, trees, water	bare-soil, buildings, cars, pavement, trees	buildings, grass, pavement, trees

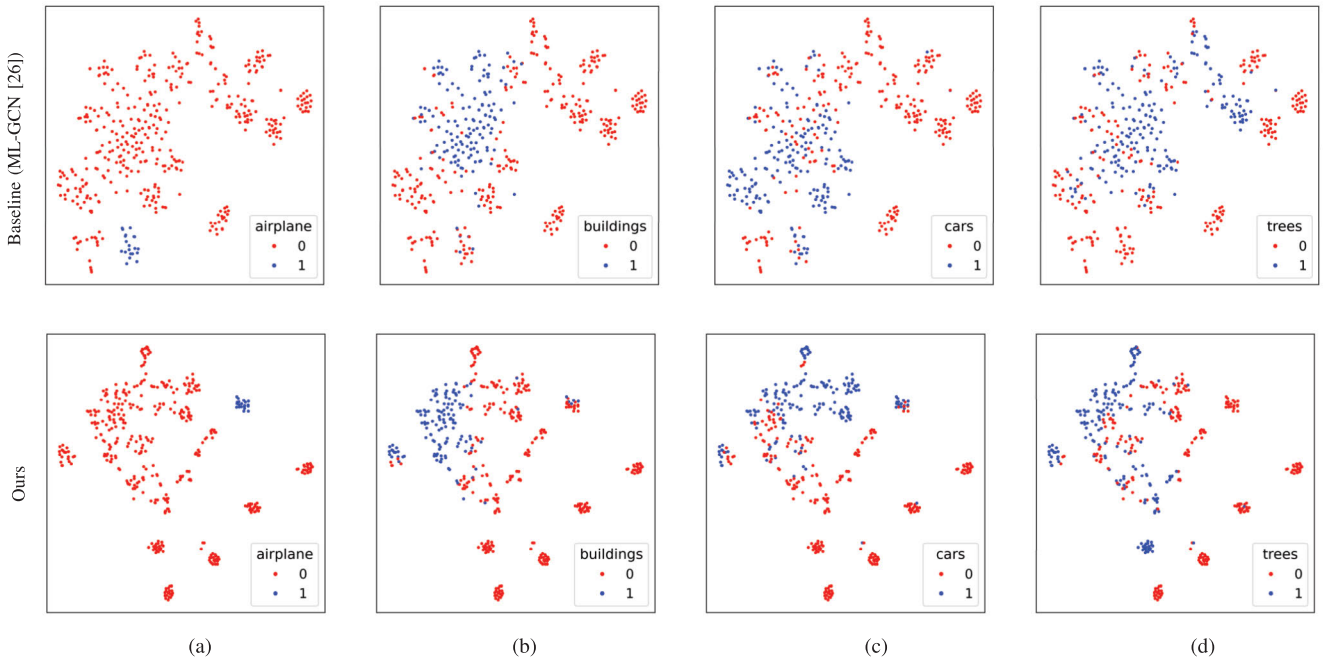


Fig. 10. t-SNE visualization of the image feature on the UCM multilabel dataset. (a) Airplane. (b) Buildings. (c) Cars. (d) Trees.

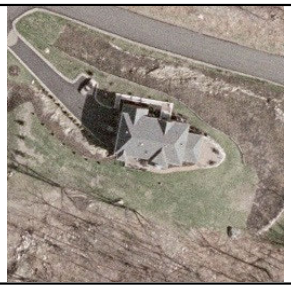
3) t-SNE Visualization: In addition, we also perform the t-SNE experiment to intuitively display the distribution of image features. Concretely, we randomly chose samples from the UCM multilabel dataset with four class labels (airplane, buildings, cars, and trees) and used t-SNE to visualize the image feature of the baseline method and the proposed method. As shown in Fig. 10, "1" represents the image global feature distribution that contains an object of the specified class, and "0" represents the opposite situation, where different colors are used to differentiate them. Overall, data points of the same class in the proposed method are tightly clustered together and have clearer boundaries compared to the baseline method. This shows that the proposed method can extract the fine-grained features of the image more efficiently through the multitask framework, the multiscale fusion strategy, and the visual attention mechanism, which helps distinguish different objects. Furthermore, it can be observed that some data points in the three classes of buildings, cars, and trees overlap in the top left part of the t-SNE visualization figure, indicating that the proposed method can enhance the semantic feature representation accuracy to effectively characterize the association between multiple objects in complex scenarios.

I. Failure Case Studies

The prediction failure cases of our proposed method are mainly divided into three types: limited number of ground-truth labels, omission of extremely small objects, and confusion of similar objects. Taking the UCM multilabel

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TABLE IX
FAILURE CASES OF OUR METHOD ON THE UCM MULTILABEL DATASET

Failure Types	Limited number of ground-truth labels	Omission of extremely small objects	Confusion of similar objects
Failure Images			
Ground Truth	cars, grass, pavement	bare-soil, buildings, cars, grass, pavement	sand, trees water
Predictions	bare-soil, cars, grass, pavement	bare-soil, buildings, grass, pavement	grass, sand, trees, water
Reasons	Lack of ground-truth labels for objects in a few images	Limited size of the model's receptive field	Confusion the features of objects in complex scenes

dataset as an example, Table IX shows the specific explanations for the above three failure cases.

1) Limited Number of Ground-Truth Labels: In multilabel remote sensing datasets, the ground-truth labels of a few images are limited, and not all objects are labeled. Because of the lack of ground-truth labels of the objects, the prediction results are considered false, despite the fact that our proposed method can accurately identify all objects in these images. As shown in Table IX, the proposed method can identify the *bare-soil*, but there is no corresponding label in the ground-truth labels. This indicates that our proposed method has a strong inference ability for multilabel prediction and can even identify objects not included in the ground-truth labels of images.

2) Omission of Extremely Small Objects: In the IFEM, due to the limited size of the model's receptive field, it is difficult to accurately extract the features of images with extremely small size. As a result, our proposed method may lead to the omission of extremely small objects, which is a common problem in multilabel remote sensing image classification. As shown in Table IX, the proposed method cannot identify the *car* due to its much smaller size than other objects in this image.

3) Confusion of Similar Objects: In the IFEM, it is difficult to distinguish the features of objects with similar colors and shapes in complex scenes, resulting in easily confusing similar objects, which is also a challenge for multilabel remote sensing image classification. As shown in Table IX, the proposed method confuses the *tree* and the *grass*.

V. CONCLUSION

In this article, we propose a multitask framework for multilabel remote sensing image classification, which consists of three modules: the IFEM, the LCEM, and the CFEM. The IFEM employs a multiscale fusion strategy and visual attention mechanism with position information to obtain high-level features of remote sensing images. The LCEM uses the graph transformer layer to improve the correlation of potentially useful label node pairs in the graph, and mines high-order correlation between label nodes using the GCN to generate more accurate label co-occurrence embedding. The CFEM establishes the class-specific feature extraction as a binary classification auxiliary task, which can improve the model's local and global feature extraction ability, to assist the main multilabel image classification task. The proposed method can efficiently obtain the fine-grained features of remote sensing images and labels. Experimental results on the UCM, AID, and DFC15 multilabel datasets show that our proposed method achieves superior performance compared with the state-of-the-art methods. In future research directions, we will further optimize our proposed method for reducing the failure cases of prediction. First, we should further explore how to extract fine-grained high-level features from remote sensing images, which provides a more fine-grained understanding of images to enhance the ability to identify extremely small objects and similar objects. Moreover, we should explore making full use of the potential correlation between labels to represent the association between multiple object-level labels in complex scenes, which further improves the performance of multilabel remote sensing image classification.

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